Speech in NLP: Text-to-Speech (TTS)

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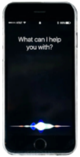
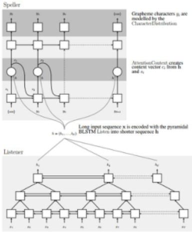
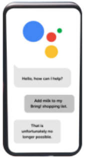
The power of human voice

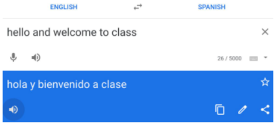
"Words mean more than what is set down on paper.
It takes the human voice to infuse them with shades of deeper meaning."
— *Maya Angelou, I Know Why the Caged Bird Sings*

Exciting Recent Developments have Disrupted the Field









2011: Apple Siri

2014: Microsoft Cortana, Amazon Alexa, Alexa Prize

2015: End-to-end neural becomes SORA

2016: Google assistant

2017: Neural TTS voice cloning

2020: Realtime speech-speech translation

Snapshot: Task Oriented Dialogue Systems

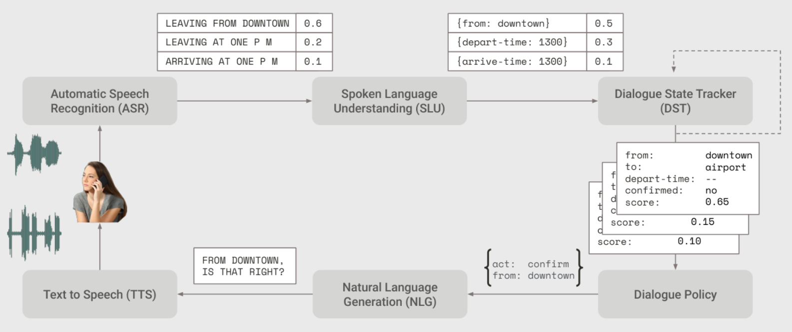


Figure 2: Architecture of dialogue-state system for task-oriented dialogue (William et al, 2016)

What is Text-to-Speech (TTS)?

- **What:** Produce speech from a text input (**Input:** text, optionally with speaker, language, style, or acoustic prompt; **Output:** waveform or codec tokens perceived as speech).
- **Applications:**
  - Accessibility (screen readers, assistive tech)
  - Personal Assistants (Apple Siri, Microsoft Cortana, Google Assistant)
  - Navigation, games, announcements/voice-overs, audiobooks, dubbing, AR/VR
  - Synthetic data for ASR

What is Text-to-Speech (TTS)?

- **Core difficulty:** Many-to-many mapping — the same text admits many natural realizations (prosody, rate, emphasis).
- **New variation:** Voice cloning — TTS systems that mimic particular speakers with minimal training data.

TTS Overview

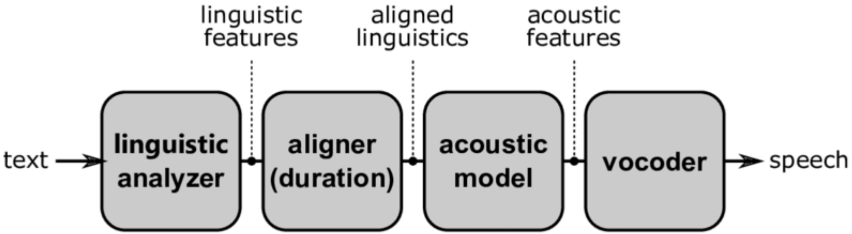
- Collect lots of speech (5–50 hours) from one speaker, transcribe very carefully, all the syllables and phones and whatnot.
- Rapid recent progress in neural approaches.
- Modern systems are DNN-based, understandable, but not yet fully emotive.
- No longer require extensive data from a single speaker to make a TTS voice for them.

TTS: Definitions

- **Codec:** A coder/decoder system that encodes analog speech signals into a digitized, discrete compressed representation for compression.
- The discrete symbols a codec produces as its compressed form can serve as **discrete codes** for language modeling.
- A codec typically consists of:
  - An **encoder** (*e.g.*, convnets) that downsamples speech into an embedding,
  - A **quantizer** that converts the embedding into a series of discrete tokens,
  - A **decoder** (*e.g.*, convnets) that upsamples the tokens/embedding back into a (lossy) reconstructed waveform.

TTS: Modular Pipeline

1. **Text analysis** — normalization, tokenization, G2P
2. **Acoustic model** — mel, latents, or codec codes over time
3. **Vocoder / decoder** — waveform or neural codec inversion



Three eras of text-to-speech

- **Pre-2016:** concatenative unit selection and statistical parametric (HMM/GMM) synthesis.
- **2016–2021:** neural end-to-end models (Tacotron 2, WaveNet vocoders, FastSpeech, Glow-TTS, VITS).
- **2022–present:** diffusion, flow matching, and **foundation audio** models (VALL-E, Voicebox) with zero-shot cloning and in-context control.
- Each era targets bottlenecks of the previous one: flexibility, fidelity, latency, then controllability at scale.

Three eras of text-to-speech

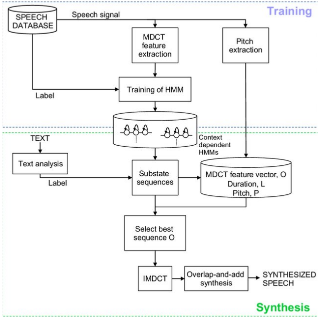
Era	**Technology**	**Strength**	**Bottleneck**
Concatenative	Unit / diphone selection	Local timbre	Huge DB; brittle joins
Statistical (SPSS)	HMM / GMM vocoder params	Small footprint; adaptation	Muffled, averaged sound
Neural (E2E)	Tacotron 2 + neural vocoder	High fidelity; simpler stack	Slow AR inference
Foundation	VALL-E, Voicebox	Zero-shot; in-context learning	Data scale; misuse risk

Pre-2016 Era: Corpus-Based and Concatenative Synthesis

- Early systems used **unit selection**: recording lots of speech from a single speaker and splitting it into small parts (like diphones or syllables).
- During synthesis, an algorithm picks which units to use by balancing:
  - **Target cost** (how well each piece matches the desired sound)
  - **Join cost** (how smooth the pieces sound when joined together)
- This gives good local naturalness, but is not very flexible and can sound glitchy if needed transitions are missing from the database.

Statistical Parametric Synthesis

- Statistical Parametric Speech Synthesis (SPSS) mainly used HMMs to model speech features instead of storing raw audio.
- SPSS models capture the spectral envelope, fundamental frequency (F_0), and duration of speech segments.
- Speaker adaptation is possible by updating a pre-trained model with a small amount of new speaker data, using MAP or linear regression techniques.
- Trajectory HMMs were introduced to make transitions between features smoother and output speech more natural.

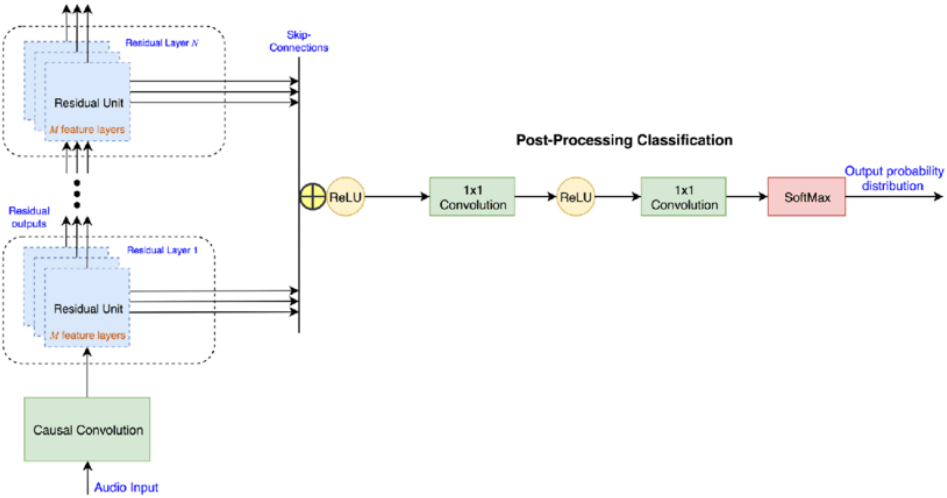


- Deep generative model for raw audio
- Autoregressive: predicts next sample given previous samples
- High-quality, natural-sounding speech
- Equation:

$$p(x) = \prod_{t=1}^{T} p(x_t | x_1, \ldots, x_{t-1})$$

- Computationally expensive for real-time use

2016–2021 Era: The Neural Revolution (WaveNet) (cont.)



Source: van den Oord et al., WaveNet, after DeepMind.

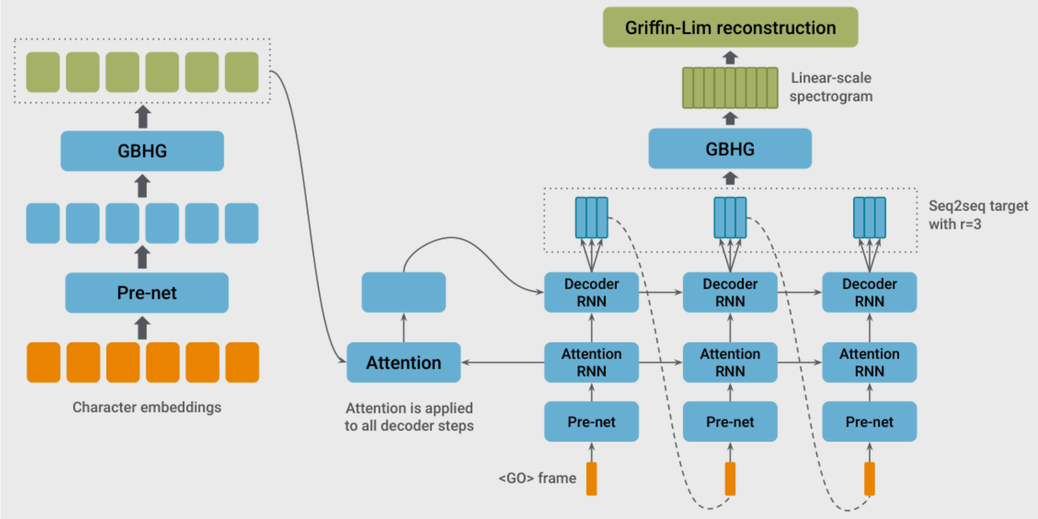
2016–2021 Era: Neural vocoders (WaveNet and beyond)

- Mel-spectrograms show how frequencies change over time, but must be turned back into sound waveforms.
- The Griffin-Lim algorithm was an early method for this, but the quality was limited.
- Neural vocoders like WaveNet can reach human-level speech quality.
- WaveNet generates raw audio samples one by one, at up to 24,000 times per second.
- It uses dilated causal convolutions, so it can "see" a lot of context efficiently.
- WaveNet models the conditional probability of each sample:

$$p(\mathbf{x} \mid \mathbf{h}) = \prod_{t=1}^{T} p(x_t \mid x_1, \ldots, x_{t-1}, \mathbf{h})$$

  where $\mathbf{h}$ is the mel-spectrogram.
- This approach gives very natural speech, but it is slow to run for real-time use.

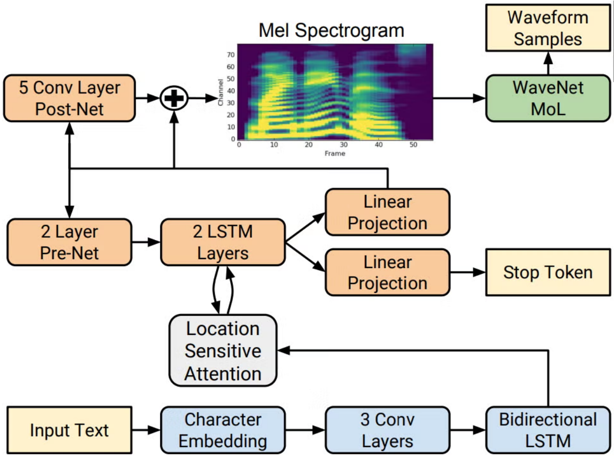
2016–2021 Era: End-to-end Neural TTS (Tacotron)



Source: Tacotron (Wang et al, 2017); Model Architecture. The model takes characters as input and outputs the corresponding

Tacotron 2

- End-to-end TTS system
- Pipeline:
  - Text $\rightarrow$ Mel-spectrogram (sequence-to-sequence model)
  - Mel-spectrogram $\rightarrow$ WaveNet/GAN vocoder (audio synthesis)
- Improved prosody and pronunciation
- Flexible for different voices and languages



Source: Tacotron 2, after Shen et al., 2018..

Autoregression: the deployment bottleneck

- Tacotron 2 + WaveNet-class vocoders raised fidelity but blocked low-latency, on-device synthesis.
- Motivated **non-autoregressive (NAR)** acoustic models and fast vocoders (FastSpeech, HiFi-GAN).

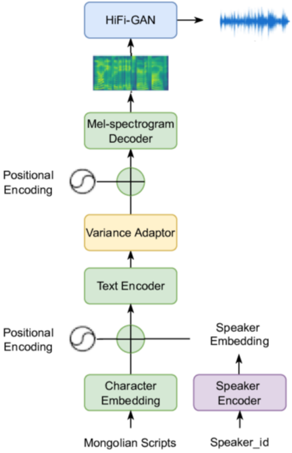
2022–present: FastSpeech, HiFi-GAN, Glow-TTS, Diffusion Models

- Non-autoregressive models: faster inference
- FastSpeech:
  - Parallel generation of spectrograms
  - Duration prediction for alignment
- HiFi-GAN:
  - GAN-based vocoder
  - High-fidelity, real-time synthesis

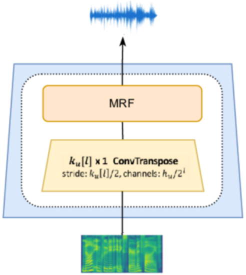
2022–present: FastSpeech, HiFi-GAN, Glow-TTS, Diffusion Models (cont.)

- Glow-TTS:
  - Flow-based generative model
  - Efficient and expressive speech synthesis
- Diffusion models:
  - Iterative refinement of audio
  - State-of-the-art naturalness

2022–present: FastSpeech, HiFi-GAN, Glow-TTS, Diffusion Models (cont.)

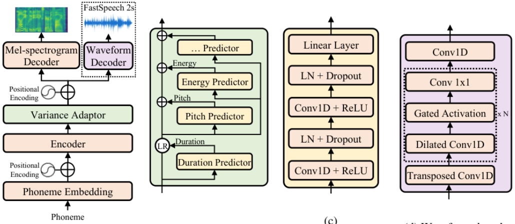


(a) Fastspeech2



(b) HiFi-GAN

- FastSpeech 2 uses a Feed-Forward Transformer (FFT) with a Length Regulator to match phoneme and spectrogram lengths, replacing soft attention.
- The Variance Adaptor predicts duration, pitch, and energy for more control over prosody.
- The Duration Predictor solves the "one-to-many" mapping, deciding exactly how many frames each phoneme lasts, which makes the system more stable.
- Pitch Predictor models the fundamental frequency (F_0); Energy Predictor models the loudness (from the spectrogram).

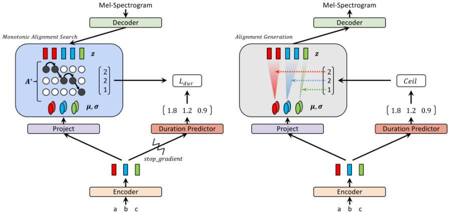


(a) FastSpeech 2 (b) Variance adaptor (c) Duration/pitch/energy predictor (d) Waveform decoder

Source: Ren et al., "FastSpeech 2: Fast and High-Quality End-to-End Text to Speech," Fig. 1(a); arXiv:2006.04558 (cropped).

Glow-TTS and monotonic alignment search (MAS)

- Glow-TTS uses Normalizing Flows to turn a simple distribution into a mel-spectrogram distribution.
- Monotonic Alignment Search (MAS) finds the single best monotonic path between text and speech using dynamic programming.
- Unlike soft attention, MAS avoids alignment breaks, so the model works well on long utterances.

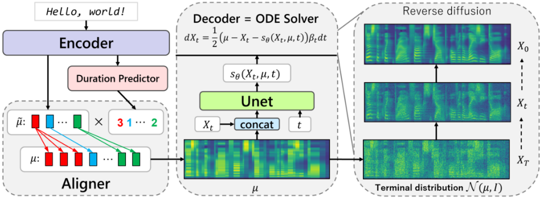


(a) An abstract diagram of the training procedure. (b) An abstract diagram of the inference procedure.

Source: Kim et al., "Glow-TTS: A Generative Flow for Text-to-Speech via Monotonic Alignment Search," Fig. 1; arXiv:2005.11129.

Grad-TTS: diffusion in the acoustic domain

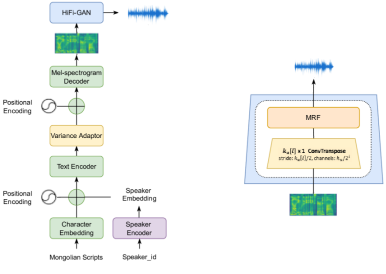
- Grad-TTS uses score-based diffusion to turn random noise into structured mel-spectrograms.
- The reverse diffusion process is controlled by stochastic differential equations (SDEs).
- Users can control the trade-off between quality and speed by choosing the number of reverse steps.
- Grad-TTS can sound good with as few as 10 steps, and it is faster and more robust than Tacotron 2.



Source: Popov et al., "Grad-TTS: A Diffusion Probabilistic Model for Text-to-Speech," Fig. 2; arXiv:2105.06337.

HiFi-GAN and parallel front ends

- HiFi-GAN is a GAN-based vocoder that can generate high-quality audio from mel-spectrograms.
- Parallel front ends like FastSpeech can generate mel-spectrograms in parallel, which is faster than sequential generation.
- The combination of FastSpeech and HiFi-GAN is a popular and effective way to synthesize speech.

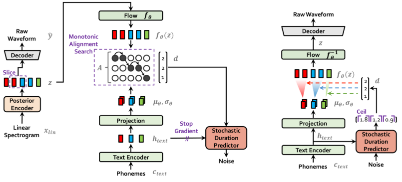


(a) Fastspeech2 (b) HiFi-GAN

Source: Kong et al., "Generative Adversarial Networks for Efficient and High Fidelity Speech Synthesis," Fig. 1; arXiv:2010.05646.

VITS: Variational Inference with Adversarial Learning for End-to-End Text-to-Speech

- Connect two modules of TTS systems through latent variables to enable efficient end-to-end learning.
  - generates more natural sounding audio than current two-stage models
- At training: a posterior encoder reads real mels; monotonic alignment search (MAS) links text to speech; a discriminator improves waveform quality.
- At inference: text $\rightarrow$ text encoder + duration predictor $\rightarrow$ latent features $\rightarrow$ waveform decoder $\rightarrow$ audio (no separate vocoder).

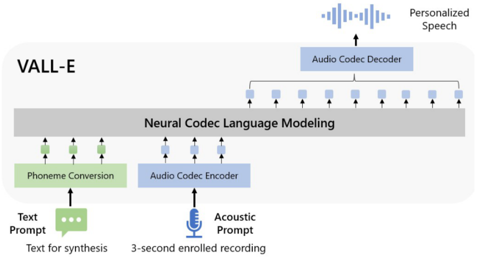


(a) Training procedure (b) Inference procedure

Source: Kim, Kong, and Son, "Conditional Variational Autoencoder with Adversarial Learning for End-to-End Text-to-Speech" (VITS), Fig. 1; arXiv:2106.06103.

VALL-E: zero-shot TTS

- VALL-E changes the TTS pipeline to phoneme → discrete code → waveform, instead of phoneme → mel-spectrogram → waveform.
- The system takes as input a text prompt and a sample of the target voice, tokenizes both (using BPE for text and an audio codec for the speech), and uses a language model to generate discrete audio tokens for the text, matching the target speaker.
- VALL-E can do zero-shot TTS, speech editing, and content creation with other generative AI models like GPT-3.



Source: Wang et al., "VALL-E: Neural Codec Language Models are Zero-Shot Text to Speech Synthesizers," Fig. 1;

	Current Systems	VALL-E
Intermediate representation	mel spectrogram	audio codec code
Objective function	continuous signal regression	language model
Training data	$\leq$ 600 hours	60K hours
In-context learning	✗	✓

Source: Chen et al., VALL-E, Fig. 2: comparison with other cascaded TTS systems.

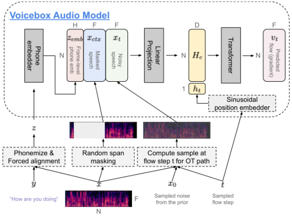
VALL-E: Limitations

- **Data requirements:** VALL-E requires large amounts of high-quality paired text and audio data for effective training.
- **Speaker similarity:** The system relies on a reference sample; quality of voice cloning can degrade for voices and accents underrepresented in training.
- **Codec/token bottlenecks:** Using discrete audio tokens (from speech codebooks) may cause artifacts or limit audio fidelity compared to raw waveform modeling.
- **Generalization:** Performance may drop for out-of-distribution prompts or novel voice/text combinations not seen in training.
- **Ethical concerns:** Zero-shot voice synthesis can enable impersonation and raises privacy and misuse risks.

Voicebox: GPT-like generalist for audio

- Non-autoregressive flow-matching model trained to fill in missing speech given context audio and full transcript.
  - Style comes from audio context; content comes from text.
- One model handles zero-shot TTS, denoising, editing, style transfer, and diverse sampling via in-context learning.
- Gains over VALL-E on intelligibility and speaker similarity, up to **20**$\times$ faster generation.

- Training: learn $p(x_{\text{masked}} \mid y, x_{\text{context}})$—fill a random masked span using transcript y and unmasked audio.
- Two modules: an audio model (80-dim log-mel frames, Transformer) and a duration model.
- **Inference:** Sample random noise, which is then shaped into meaningful speech representations (mel spectrogram frames) by solving a continuous flow ordinary differential equation (ODE) conditioned on both the transcript and audio context. After obtaining the predicted mel spectrogram, a neural vocoder is used to convert these mel frames into audible waveform speech, thus completing the synthesis pipeline.



Source: Le et al., Voicebox, Fig. 1: task generalization through in-context learning; arXiv:2306.15687.

Voicebox: limitations

- **Read speech domain:** trained mainly on audiobooks; may not transfer well to casual conversation (laughter, back-channels).
- **Front-end dependency:** needs a phonemizer and forced aligner; word-based phonemizers miss context-sensitive pronunciation.
- **Limited control:** can transfer overall style from prompts, but cannot separately pick voice from one clip and emotion from another.
- **Bias risk:** performance can drop for underrepresented accents unless training data stays diverse.
- **Misuse:** high-quality cloning raises deepfake concerns—authors release a strong synthetic-speech **classifier** as a mitigation.

Architectures at a glance

Model	**Paradigm**	**Representation**	**Highlight**
Tacotron 2	AR seq2seq	Mel	Strong naturalness
FastSpeech 2	NAR Transformer	Mel	Fast; explicit prosody
Glow-TTS	Flow + MAS	Mel	Hard alignments
VALL-E	Codec LM	Discrete codes	Zero-shot cloning
Voicebox	Flow matching	Continuous audio	Infilling / editing

Applications

- **Machine learning applications**
  - Extract information from speech using supervised learning
  - Emotion, speaker ID, flirtation, deception, depression, intoxication
- **Dialog system / SLU applications**
  - Building systems to solve a problem
  - Medical transcription, reservations via chat
- **New area: Self-supervised foundation models**

Evaluating TTS quality

- **Human tests:** Does the voice sound natural?
- **Similarity:** Is the generated speech close to real speech?
- **Intelligibility:** Can speech-to-text systems understand it?
- **Cloning:** How well does it copy a specific voice?
- TTS is evaluated by playing a sentence to human listeners and having them give a mean opinion score (MOS) on a scale of 1 to 5.

Ethics before you collect speech

Don't record someone without their consent

In California, all parties to any confidential conversation must give their consent to be recorded. For calls occurring over cellular or cordless phones, all parties must consent before a person can record, regardless of confidentiality.

Don't create a speech synthesizer / voice clone of someone without their consent

It might be fun, but it's a little creepy. People get upset.

Okay to use existing speech datasets (we'll provide some).

Do consider subgroup and language bias when building systems

Poor performance on subgroups e.g. non-native speakers

Many languages are under-served relative to English/Mandarin.

Source: Stanford CS224S, Lecture 1 (2024 Spring).

Takeaways

- One text can be spoken many ways (speed, stress, intonation). Most systems still go text $\rightarrow$ acoustic features $\rightarrow$ waveform; newer models like VALL-E predict audio tokens directly.
- TTS improved in stages: stitch recorded clips $\rightarrow$ slow but high-quality neural models (Tacotron 2, WaveNet) $\rightarrow$ faster parallel models (FastSpeech 2, HiFi-GAN, VITS) $\rightarrow$ foundation models that clone voices from a short sample (VALL-E, Voicebox).
- Voice cloning from a few seconds of audio works well today, but can fail on rare accents; use it carefully with consent, labeling, and fake-speech detection.

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